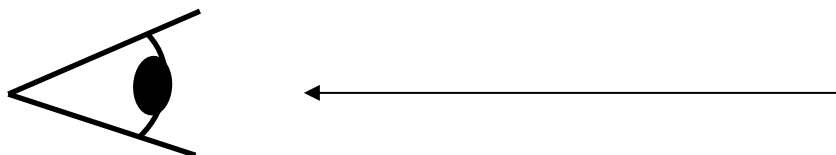


What is light?

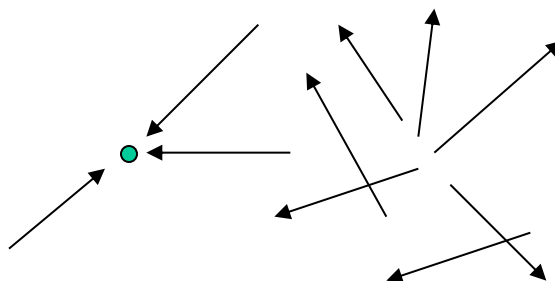
Electromagnetic radiation (EMR) moving along rays in space

- $R(\lambda)$ is EMR, measured in units of power (watts)
 - λ is wavelength



Light field

- We can describe all of the light in the scene by specifying the radiation (or “**radiance**” along all light rays) arriving at every point in space and from every direction



$$R(X, Y, Z, \theta, \phi, \lambda, t)$$

The light field

$$R(X, Y, Z, \theta, \phi, \lambda, t)$$

- Known as the **plenoptic function**
- If you know R , you can predict how the scene would appear from any viewpoint.

Additional Reading

The plenoptic function

- **Original reference:** E. Adelson and J. Bergen, "[The Plenoptic Function and the Elements of Early Vision](#)," in M. Landy and J. A. Movshon, (eds) Computational Models of Visual Processing, MIT Press 1991.
- L. McMillan and G. Bishop, "[Plenoptic Modeling: An Image-Based Rendering System](#)", Proc. SIGGRAPH, 1995, pp. 39-46.

The light field

- M. Levoy and P. Hanrahan, "[Light Field Rendering](#)", Proc SIGGRAPH 96, pp. 31-42.
- S. J. Gortler, R. Grzeszczuk, R. Szeliski, and M. F. Cohen, "[The lumigraph](#)," in Proc. SIGGRAPH, 1996, pp. 43-54.

The interaction of light and matter

What happens when a light ray hits a point on an object?

- Some of the light gets absorbed
 - converted to other forms of energy (e.g., heat)
- Some gets transmitted through the object
 - possibly bent, through “refraction”
- Some gets reflected
 - as we saw before, it could be reflected in multiple directions at once

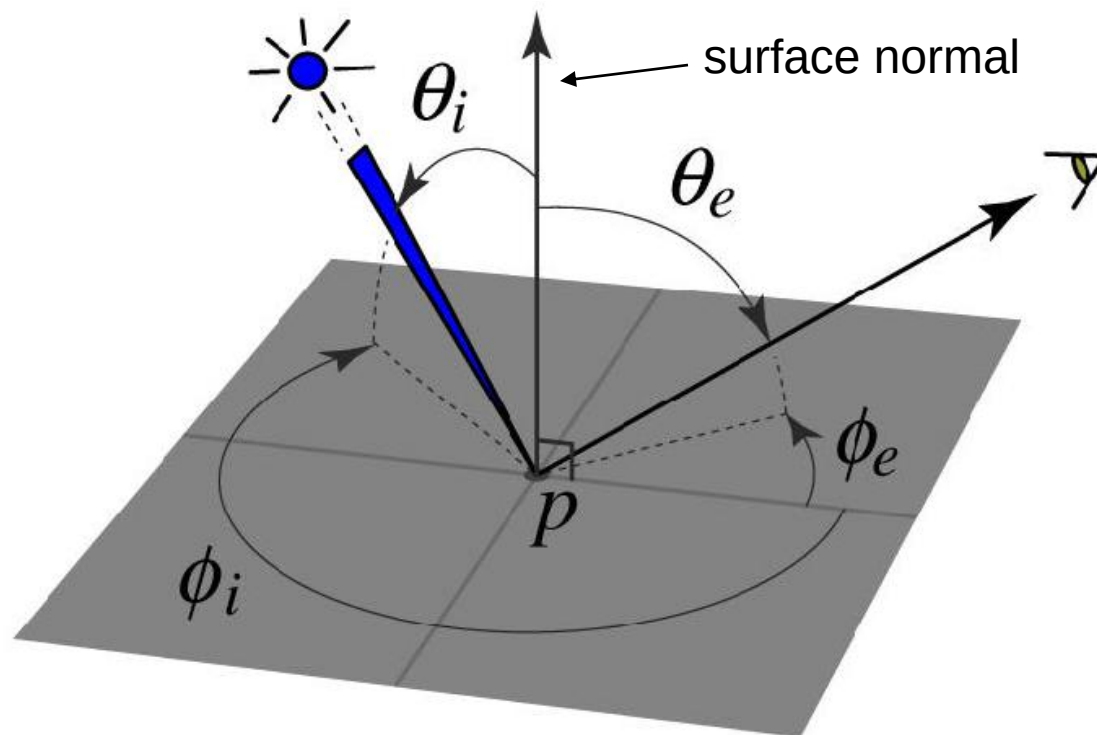
Let's consider the case of reflection in detail

- In the most general case, a single incoming ray could be reflected in all directions. How can we describe the amount of light reflected in each direction?

The BRDF

The Bidirectional Reflection Distribution Function

- Given an incoming ray (θ_i, ϕ_i) and outgoing ray (θ_e, ϕ_e)
what proportion of the incoming light is reflected along outgoing ray?

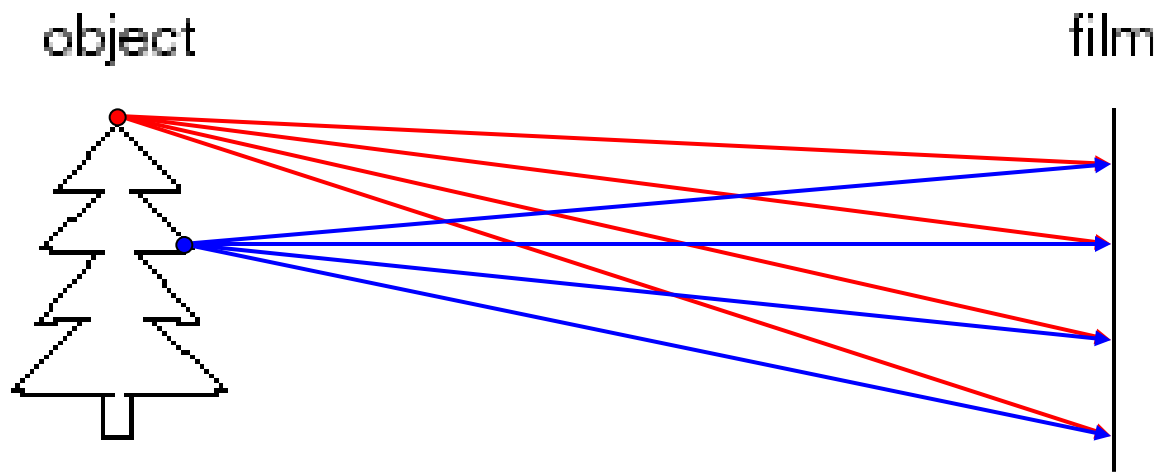


Answer given by the BRDF: $\rho(\theta_i, \phi_i, \theta_e, \phi_e)$

We will first study the geometry of image formation.

Later, we will look into illumination effects.

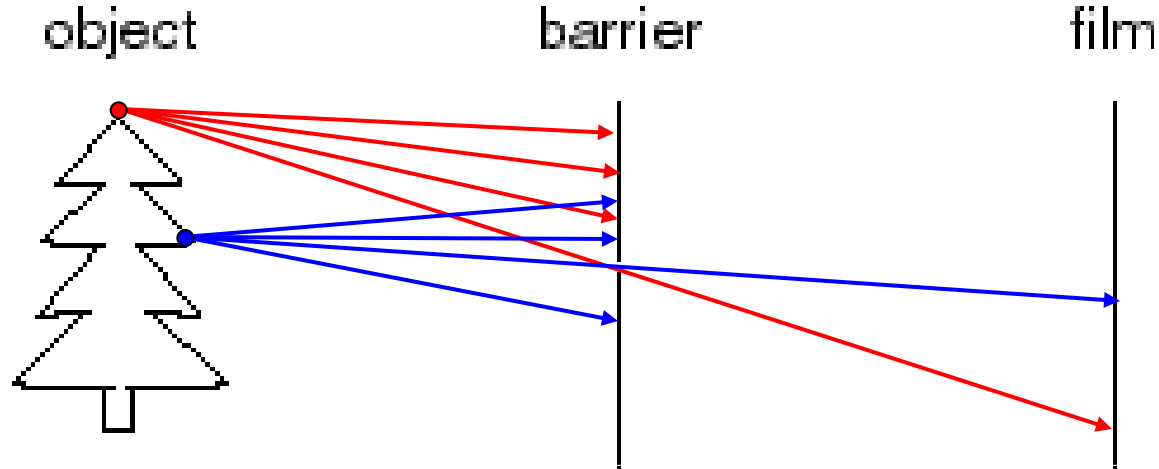
Image formation



Let's design a camera

- Idea 1: put a piece of film in front of an object
- Do we get a reasonable image?

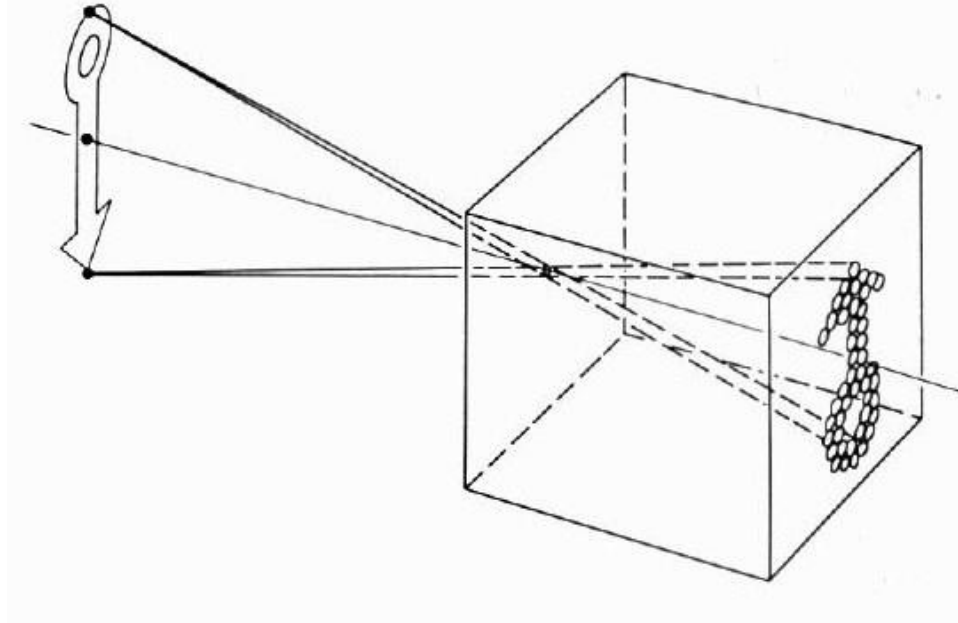
Pinhole camera



Add a barrier to block off most of the rays

- This reduces blurring
- The opening known as the **aperture**
- How does this transform the image?

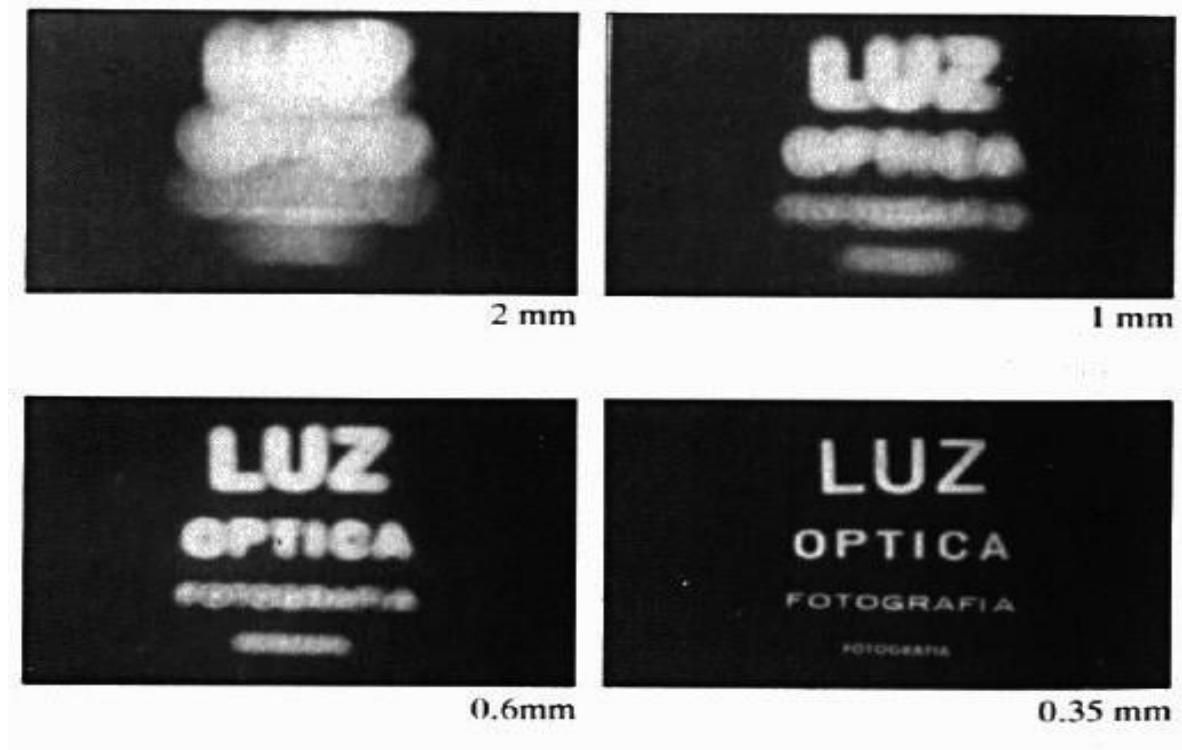
Camera Obscura



The first camera

- Known to Aristotle
- How does the aperture size affect the image?

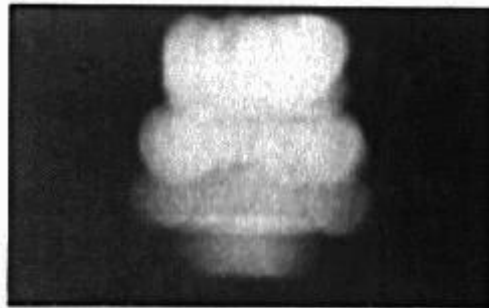
Shrinking the aperture



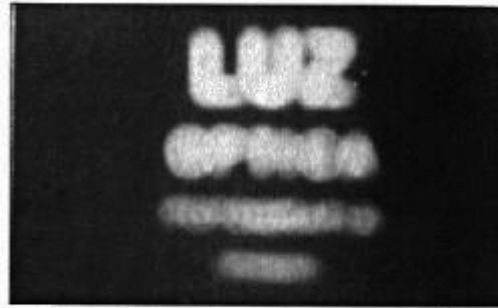
Why not make the aperture as small as possible?

- Less light gets through
- *Diffraction* effects...

Shrinking the aperture



2 mm



1 mm



0.6mm



0.35 mm

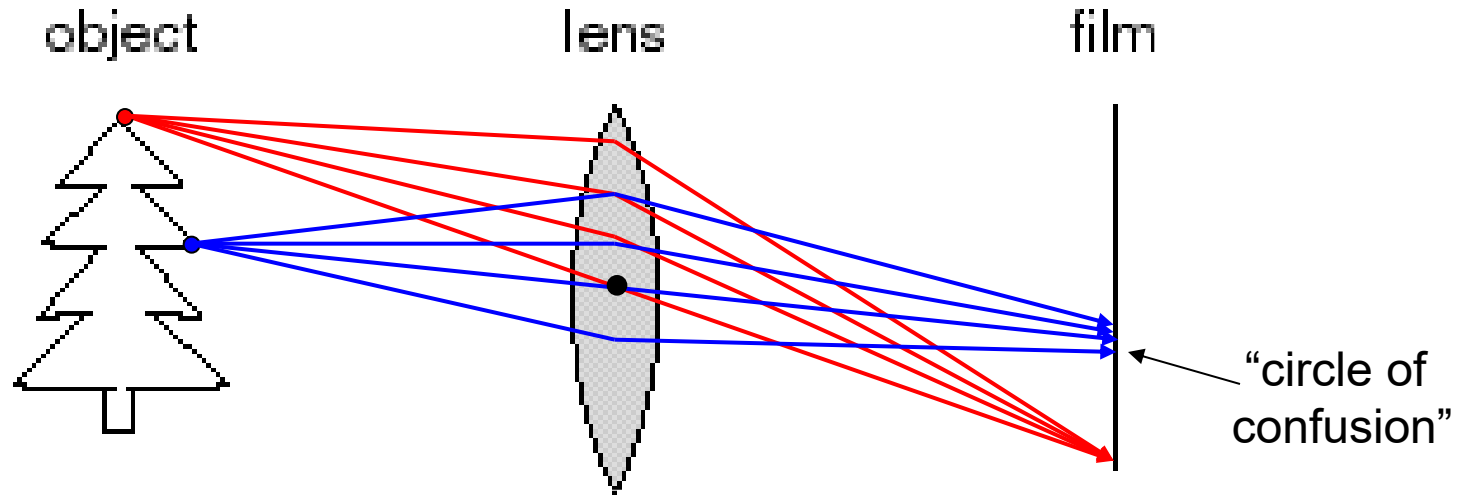


0.15 mm



0.07 mm

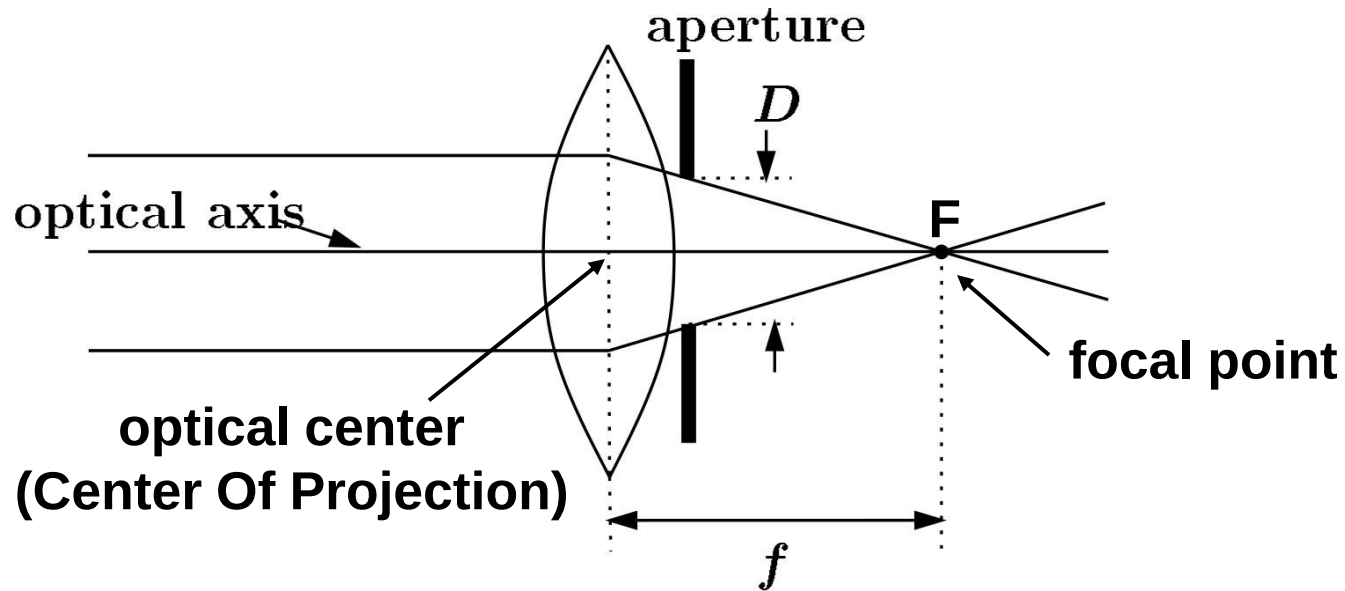
Adding a lens



A lens focuses light onto the film

- There is a specific distance at which objects are “in focus”
 - other points project to a “circle of confusion” in the image
- Changing the shape of the lens changes this distance

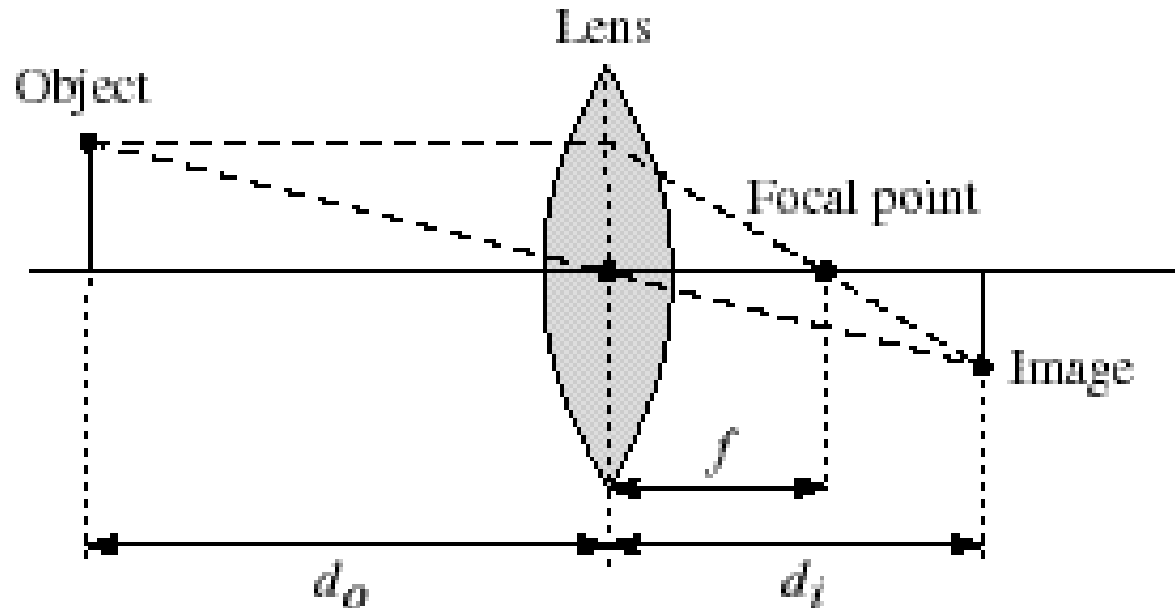
Lenses



A lens focuses parallel rays onto a single focal point

- focal point at a distance f beyond the plane of the lens
 - f is a function of the shape and index of refraction of the lens
- Aperture of diameter D restricts the range of rays
 - aperture may be on either side of the lens
- Lenses are typically spherical (easier to produce)

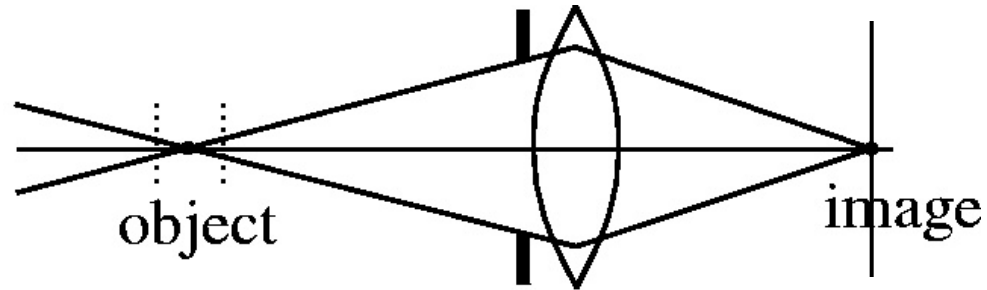
Thin lenses



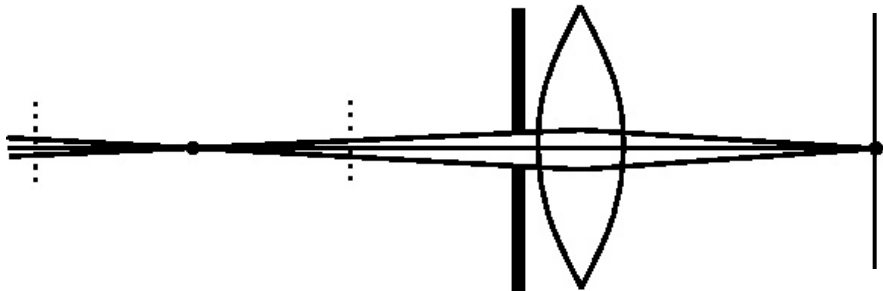
Thin lens equation:
$$\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}$$

- Any object point satisfying this equation is in focus
- What is the shape of the focus region?
- How can we change the focus region?
- Thin lens applet: http://www.phy.ntnu.edu.tw/java/Lens/lens_e.html (by Fu-Kwun Hwang)

Depth of field



$f/5.6$

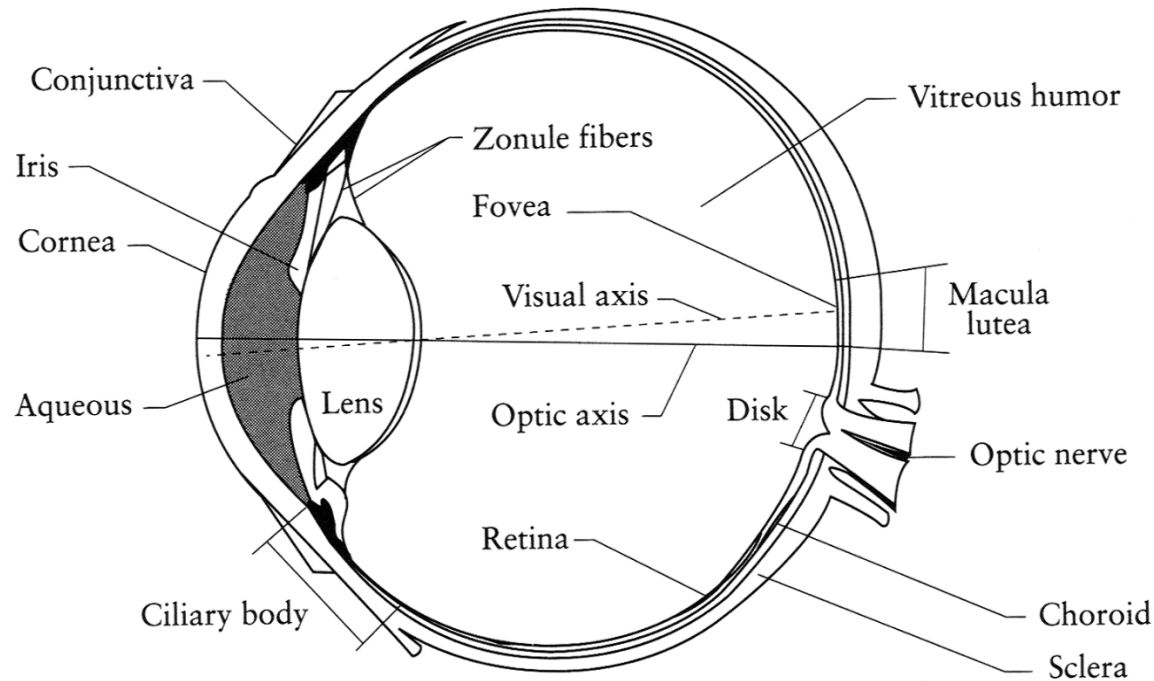


$f/32$

Changing the aperture size affects depth of field

- A smaller aperture increases the range in which the object is approximately in focus

The eye



The human eye is a camera

- **Iris** - colored annulus with radial muscles
- **Pupil** - the hole (aperture) whose size is controlled by the iris
- What's the "film"?
 - photoreceptor cells (rods and cones) in the **retina**

Digital camera



A digital camera replaces film with a sensor array

- Each cell in the array is a **Charge Coupled Device**
 - light-sensitive diode that converts photons to electrons
 - other variants exist: CMOS is becoming more popular
 - CMOS: power-efficient, cheaper; CCD: higher image quality

Issues with digital cameras

Noise

- big difference between consumer vs. SLR-style cameras
- low light is where you most notice [noise](#)

Compression

- creates [artifacts](#) except in uncompressed formats (tiff, raw)

Color

- [color fringing](#) artifacts from [Bayer patterns](#)

Blooming

- charge [overflowing](#) into neighboring pixels

In-camera processing

- oversharpening can produce [halos](#)

Interlaced vs. progressive scan video

- [even/odd rows from different exposures](#)

Are more megapixels better?

- requires higher quality lens
- noise issues

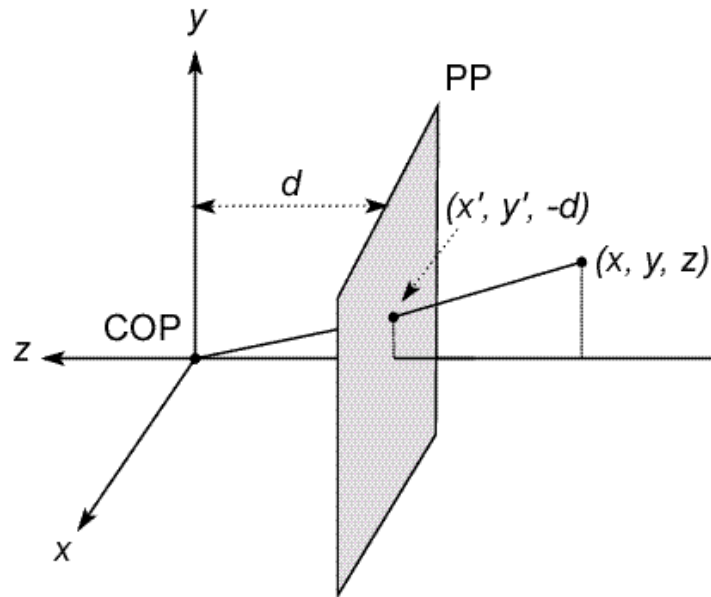
Stabilization

- compensate for camera shake (mechanical vs. electronic)

More info online, e.g.,

- <http://electronics.howstuffworks.com/digital-camera.htm>
- <http://www.dpreview.com/>

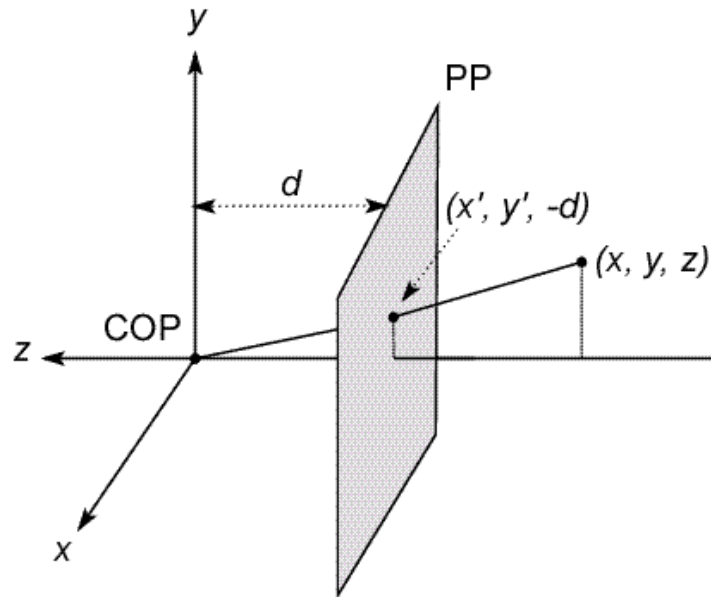
Modeling projection



The coordinate system

- We will use the pin-hole model as an approximation
- Put the optical center (**C**enter **O**f **P**rojection) at the origin
- Put the image plane (**P**rojection **P**lane) *in front* of the CO
- The camera looks down the *negative* z axis
 - we need this if we want right-handed-coordinates

Modeling projection



Projection equations

- Compute intersection with PP of ray from (x,y,z) to COP
- Derived using similar triangles

$$(x, y, z) \rightarrow \left(-d\frac{x}{z}, -d\frac{y}{z}, -d\right)$$

- We get the projection by throwing out the last coordinate:

$$(x, y, z) \rightarrow \left(-d\frac{x}{z}, -d\frac{y}{z}\right)$$

Homogeneous coordinates

Is this a linear transformation?

- no—division by z is nonlinear

Trick: add one more coordinate:

$$(x, y) \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

homogeneous image
coordinates

$$(x, y, z) \Rightarrow \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

homogeneous scene
coordinates

Converting *from* homogeneous coordinates

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} \Rightarrow (x/w, y/w)$$

$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \Rightarrow (x/w, y/w, z/w)$$

Perspective Projection

Projection is a matrix multiply using homogeneous coordinates:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1/d & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ -z/d \end{bmatrix} \Rightarrow \left(-d\frac{x}{z}, -d\frac{y}{z} \right)$$

divide by third coordinate

This is known as **perspective projection**

- The matrix is the **projection matrix**

Perspective Projection

How does scaling the projection matrix change the transformation?

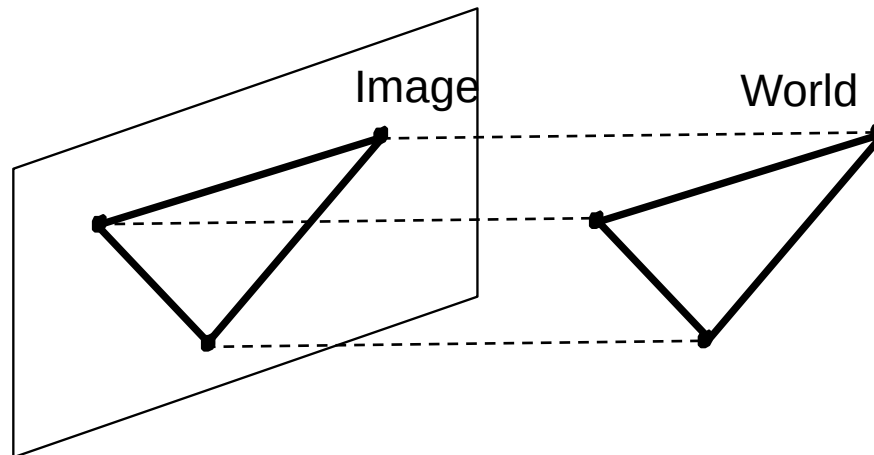
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1/d & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ -z/d \end{bmatrix} \Rightarrow \left(-d\frac{x}{z}, -d\frac{y}{z}\right)$$

$$\begin{bmatrix} -d & 0 & 0 & 0 \\ 0 & -d & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} -dx \\ -dy \\ z \end{bmatrix} \Rightarrow \left(-d\frac{x}{z}, -d\frac{y}{z}\right)$$

Orthographic projection

Special case of perspective projection

- Distance from the COP to the PP is infinite



- Good approximation for telephoto optics
- Also called “parallel projection”: $(x, y, z) \rightarrow (x, y)$
- What’s the projection matrix?

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \Rightarrow (x, y)$$

Variants of orthographic projection

Scaled orthographic

- Also called “weak perspective”

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1/d \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ 1/d \end{bmatrix} \Rightarrow (dx, dy)$$

Affine projection

- Also called “paraperspective”

$$\begin{bmatrix} a & b & c & d \\ e & f & g & h \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

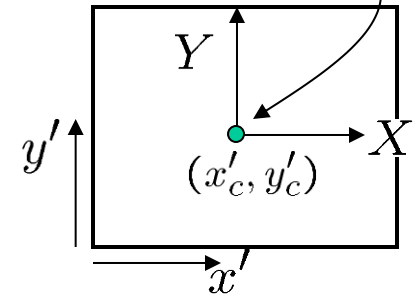
Camera parameters

A camera is described by several parameters

- Translation **T** of the optical center from the origin of world coords
- Rotation **R** of the image plane
- focal length **f**, principle point (x'_c, y'_c) , pixel size (s_x, s_y)
- blue parameters are called “**extrinsics**,” red are “**intrinsics**”

Projection equation

$$\mathbf{X} = \begin{bmatrix} sx \\ sy \\ s \end{bmatrix} = \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \mathbf{\Pi} \mathbf{X}$$



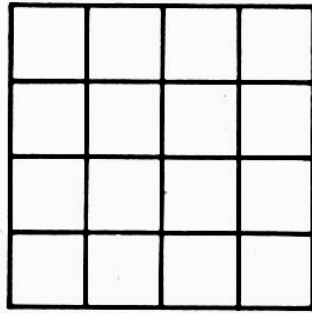
- The projection matrix models the cumulative effect of all parameters
- Useful to decompose into a series of operations

$$\mathbf{\Pi} = \underbrace{\begin{bmatrix} -fs_x & 0 & x'_c \\ 0 & -fs_y & y'_c \\ 0 & 0 & 1 \end{bmatrix}}_{\text{intrinsics}} \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}}_{\text{projection}} \underbrace{\begin{bmatrix} \mathbf{R}_{3 \times 3} & \mathbf{0}_{3 \times 1} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix}}_{\text{rotation}} \underbrace{\begin{bmatrix} \mathbf{I}_{3 \times 3} & \mathbf{T}_{3 \times 1} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix}}_{\text{translation}}$$

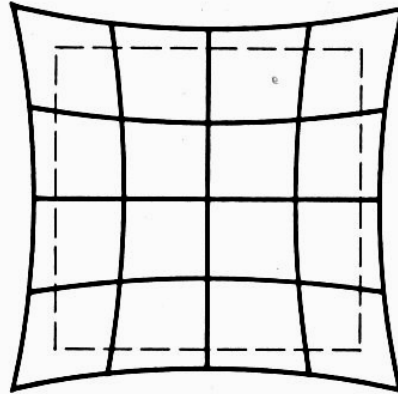
identity matrix

- The definitions of these parameters are **not** completely standardized
 - especially intrinsics—varies from one book to another

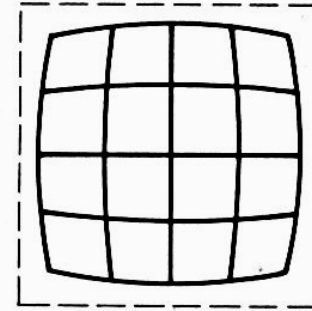
Distortion



No distortion



Pin cushion



Barrel

Radial distortion of the image

- Caused by imperfect lenses
- Deviations are most noticeable for rays that pass through the edge of the lens

Correcting radial distortion



from [Helmut Dersch](#)

Modeling distortion

Project $(\hat{x}, \hat{y}, \hat{z})$
to “normalized”
image coordinates

$$x'_n = \hat{x} / \hat{z}$$
$$y'_n = \hat{y} / \hat{z}$$

Apply radial distortion

$$r^2 = x_n'^2 + y_n'^2$$
$$x'_d = x'_n(1 + \kappa_1 r^2 + \kappa_2 r^4)$$
$$y'_d = y'_n(1 + \kappa_1 r^2 + \kappa_2 r^4)$$

Apply focal length
translate image center

$$x' = f x'_d + x_c$$
$$y' = f y'_d + y_c$$

To model lens distortion

- Use above projection operation instead of standard projection matrix multiplication

360 degree field of view...



Basic approach

- Take a photo of a parabolic mirror with an orthographic lens (Nayar)
- Or buy one a lens from a variety of omnicam manufacturers...
 - See <http://www.cis.upenn.edu/~kostas/omni.html>